unaware of the others' existence. But if we follow the rules, we can calculate what happens using particle language even when a bunch of fields are vibrating right on top of one another, exactly when you might expect their field-ness to be most important.

That's the essential insight from Richard Feynman and his well-known tool of *Feynman diagrams*. When he first invented his diagrams, Feynman held out the hope that he was suggesting a particle-based alternative to quantum field theory, but that turns out not to be the case. What they are is both a wonderfully vivid metaphorical device and an incredibly convenient computational method, within the overarching paradigm of quantum field theory.

A Feynman diagram is simply a stick-figure cartoon representing particles moving and interacting with one another. With time running from left to right, an initial set of particles comes in, they jumble up with various particles appearing or disappearing, then a final set of particles emerges. Physicists use these diagrams not only to describe what processes are allowed to happen but to precisely calculate the likelihood that they actually will. If you want to ask, for example, what particles a Higgs boson might decay into and how rapidly, you would do a calculation involving a boatload of Feynman diagrams, each representing a certain contribution to the final answer. Likewise if you want to know how likely it is that an electron and a positron will scatter off each other.

Here is a simple Feynman diagram. The way to think about this picture is that an electron and a positron (straight lines) come in from the left, meet each other, and annihilate into a photon (wavy line), which travels for a while before converting back into an electron/positron pair. There are specific rules that allow physicists to attach precise numbers to every such diagram, indicating the contribution that this picture makes to the overall process of "an electron and a positron scatter off each other."